

How COVID Reshaped the Efficient Frontier: Evidence from Indian Equities

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Abstract

This paper investigates whether the mean-variance efficient frontier for Indian equities shifted meaningfully between the pre-COVID period (January 2015–January 2020) and the post-COVID period (July 2021–December 2024). Using 45 continuously-listed stocks from the NIFTY 50 as of January 2015, daily and weekly log returns, Ledoit-Wolf shrinkage covariance estimation, and long-only constrained quadratic programming, we construct and compare efficient frontiers across both regimes. The post-COVID frontier lies strictly above and to the right of the pre-COVID frontier at both return frequencies, with the tangency portfolio Sharpe ratio improving from 1.02 to 1.93. Frontier decomposition shows that most of this improvement is attributable to higher expected returns (μ) rather than changes in the covariance structure (Σ). The frontier shift is accompanied by a sectoral rotation where financials were displaced by pharma and telecom. Portfolio concentration also declined, with the effective number of positions rising from 3.9 to 6.4 (measured using HHI). We interpret these findings through the lens of RBI monetary accommodation, PLI scheme-driven industrial re-rating, post-COVID retail investor participation, and the China+1 FII inflow dynamic.

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1 Introduction

The COVID-19 pandemic was one of the most severe and sudden macroeconomic disruptions in modern history. For equity investors, it compressed into a matter of weeks what would ordinarily unfold over years: a sharp crash, an unprecedented policy response, a recovery, and the emergence of a structurally different economic landscape. In India, the NIFTY 50 fell approximately 26% in six weeks between late February and late March 2020, before recovering to new all-time highs by mid-2021 (Figure 1).

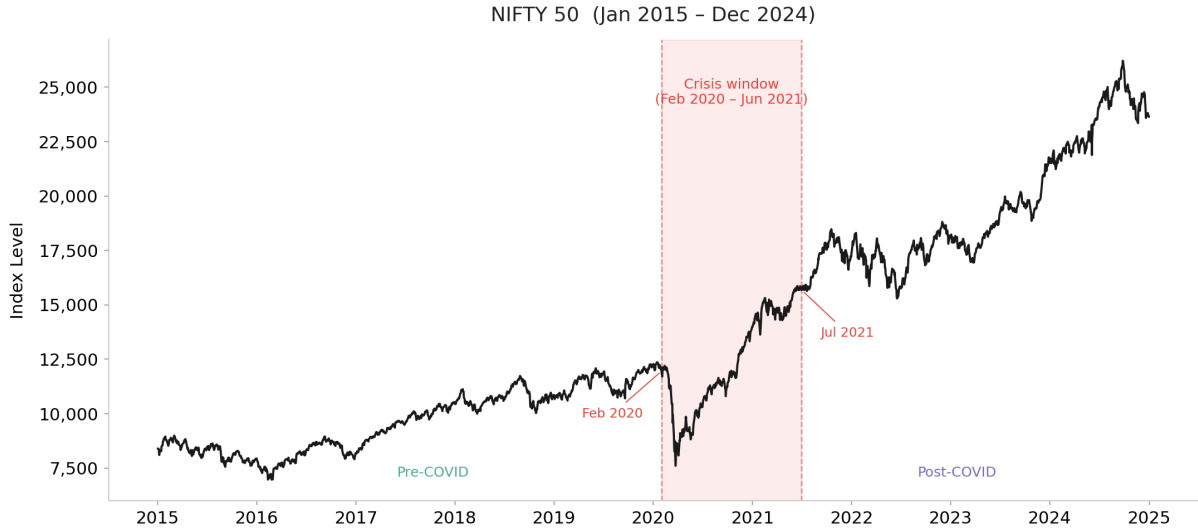


Figure 1: NIFTY 50

Against this backdrop, an important question emerges: did COVID-19 alter the investment opportunity set available to Indian equity investors in a persistent way? Put differently, did the pandemic change the nature of the risk-return trade-off itself, rather than simply generating a temporary market shock?

In the mean-variance framework of Markowitz [1952], the efficient frontier represents the set of portfolios that deliver the highest expected return for a given level of risk. Any structural shift in this frontier therefore implies that the menu of feasible risk-return combinations available to investors has changed. Such a shift would affect portfolio allocation decisions across the entire spectrum of investor risk preferences. India presents a compelling setting for this investigation for several reasons. First, the structural forces that accompanied COVID-19 in India were unusually concentrated and policy-driven: the Reserve Bank of India cut the repo rate from 5.15% to 4.0%, the government launched Production-Linked Incentive (PLI) schemes that materially altered the earnings trajectory of capital goods and pharmaceutical companies, and retail investor participation surged from 4 crore Demat accounts in 2019 to over 14 crore by 2024. Second, India's post-COVID growth trajectory diverged sharply from developed markets, attracting Foreign

Institutional Investor (FII) inflows as global supply chains adopted a China+1 strategy. Third, the sectoral composition of Indian equities shifted in ways that were both rapid and persistent— particularly with the rise of pharma, industrials, and telecom alongside the relative stagnation of the financial sector. This paper addresses the research question directly:

Does the mean-variance efficient frontier for Indian equities shift meaningfully between the pre-COVID (2015–2020) and post-COVID (2021–2024)?

We find that the evidence points toward a substantial economic shift in the post-COVID investment opportunity set. The post-COVID frontier lies strictly above and to the right of the pre-COVID frontier across the full range of achievable volatility levels, at both daily and weekly return frequencies. The tangency portfolio Sharpe ratio improved by +89% on a daily basis. Frontier decomposition shows that over 95% of this improvement is attributable to higher expected returns rather than changes in the covariance structure. Box’s M test formally confirms that the covariance matrix changed structurally ($p = 3 \times 10^{-6}$). Sector weight analysis reveals a portfolio rotation, with financials exiting the optimal portfolio and pharma and telecom entering from zero. The paper is structured as follows. Section 2 reviews the relevant literature. Section 3 describes the dataset and its sources. Section 4 presents exploratory data analysis. Section 5 outlines the methodology. Section 7 describes robustness checks. Section 6 presents the core results. Section 8 concludes.

2 Literature Review

2.1 Mean-Variance Optimisation and Its Limitations

Markowitz [1952] introduced the mean-variance framework, establishing that a rational investor should care only about the expected return and variance of a portfolio, and that the efficient set of portfolios — the efficient frontier — is traced by minimising variance for each target return level. Sharpe [1964] extended this via the Capital Asset Pricing Model to derive the Capital Allocation Line and the concept of the tangency portfolio. A well-documented limitation of the classical framework is sensitivity to estimation error in input parameters. Merton [1980] showed that even with long samples, mean return estimation is imprecise. Michaud [1989] termed this “error maximisation”: the optimiser concentrates weight in assets with the highest estimated mean returns, amplifying estimation errors. Black & Litterman [1992] proposed the Views framework as one response; Ledoit & Wolf [2004] proposed covariance shrinkage as another, which we adopt in this study. DeMiguel et al. [2009] showed that naive $1/N$ portfolios often outperform optimised portfolios out-of-sample, motivating the use of shrinkage and the long-only constraint as mitigants.

2.2 Spanning and Frontier Shift Tests

The statistical question of whether one frontier is significantly different from another is addressed by the mean-variance spanning literature. Huberman & Kandel [1987] formalised the spanning test: a set of new assets spans the existing opportunity set if and only if adding those assets does not move the efficient frontier. Jobson & Korkie [1989] proposed an asymptotic z -test for the equality of two Sharpe ratios; Memmel [2003] corrected an algebraic error in the original variance formula that becomes material in small samples. Ledoit and Wolf [2008] proposed a distribution-free bootstrap version that relaxes the normality and i.i.d. assumptions. Kan & Zhou [2012] provide a comprehensive survey of spanning tests and their power properties.

2.3 Indian Equity Markets and COVID-19

The Indian equity market has attracted growing academic attention. Akhtaruzzaman et al. [2021] examine financial contagion during COVID-19 and find substantial changes in correlation structure across emerging market equities. Ang & Bekaert [2002] develop a regime-switching model showing correlations are systematically higher in bear-market regimes. Longin & Solnik [2001] confirm that international equity correlations spike in falling markets. Studies examining pre- versus post-COVID portfolio efficiency have generally focused on correlation structure changes rather than the full frontier. Our contribution is to apply a rigorous mean-variance framework specifically to Indian equities with a clean experimental design that isolates the two regimes from the COVID crash itself.

2.4 Covariance Estimation

Ledoit & Wolf [2004] demonstrated analytically that shrinking the sample covariance matrix toward a structured target reduces estimation error under the Frobenius norm and substantially improves the stability and out-of-sample performance of mean-variance portfolios, particularly in settings with many assets relative to the sample size. Jagannathan & Ma [2003] showed that imposing a long-only constraint acts as a form of implicit regularisation by restricting extreme portfolio weights and preventing the optimiser from exploiting unstable or spurious negative correlations present in the sample covariance matrix. Font [2016] developed a bootstrap-based framework for constructing statistical confidence bands around the estimated efficient frontier, allowing researchers to assess whether observed differences in frontier locations are economically meaningful or simply driven by sampling variation.

3 Dataset: Sources and Description

3.1 Asset Universe

The asset universe consists of 45 stocks selected from the NIFTY 50 index constituents as of January 1, 2015. We begin from the NIFTY 50 because it represents the 50 largest and most liquid Indian equities by free-float market capitalisation, ensuring data quality and implementability. Five stocks are excluded for data integrity reasons: HDFC Ltd (merged into HDFC Bank in July 2023, truncating the post-COVID estimation window), Cairn India (merged into Vedanta in April 2017), Tata Motors (discontinuous price history on Yahoo Finance), and two others with insufficient continuous data. The universe is held fixed across both estimation periods — the same 45 stocks are used pre- and post-COVID. This deliberate choice ensures we test whether the frontier defined by a fixed opportunity set changed, not whether different assets look better in different periods.

3.2 Price Data and Returns

Daily adjusted closing prices are sourced from Yahoo Finance via the `yfinance` Python library, using the `.NS` suffix for NSE-listed securities. Yahoo Finance provides dividend- and split-adjusted prices, so computed returns represent total equity returns including reinvested dividends. We compute *log returns* throughout the analysis:

$$r_{i,t} = \ln\left(\frac{P_{i,t}}{P_{i,t-1}}\right) \quad (1)$$

Log returns are preferred over simple returns for three reasons: they are time-additive (monthly returns can be aggregated from daily by summation), they are better approximated by a Gaussian distribution for short intervals, and they are the standard in portfolio theory with continuously compounded returns. Annualisation is performed by scaling by $k = 252$ (daily) or $k = 52$ (weekly). The use of both daily and weekly frequencies serves as a primary robustness check. Daily returns can be contaminated by bid-ask bounce and non-synchronous trading effects among less liquid stocks. Weekly returns average over these microstructure artefacts. Consistent findings across both frequencies provide stronger evidence of genuine structural change.

3.3 Estimation Periods

Table 1: Estimation periods and key features

Period	Window	Obs. (Daily)	Key Features
Pre-COVID	Jan 2015 – Jan 2020	$\approx 1,251$	Demonetisation (2016), GST (2017), IL&FS crisis (2018)
Crisis (excluded)	Feb 2020 – Jun 2021	—	COVID crash, RBI emergency cuts, recovery rally
Post-COVID	Jul 2021 – Dec 2024	≈ 865	PLI schemes, retail boom, China+1 flows, rate hike cycle

The exclusion of the crisis window (February 2020 – June 2021) is central to our empirical design. Periods of acute financial stress are characterised by a sharp increase in cross-asset correlations as investors engage in broad-based liquidation and flight-to-cash behaviour, reducing the benefits of diversification. This phenomenon of “correlation breakdown” or “correlation contagion” during crises is well documented in the literature, including Ang & Bekaert [2002] and Longin & Solnik [2001]. Including the crash period directly in either regime would therefore risk conflating temporary panic-driven market dynamics with more persistent structural changes in the investment opportunity set.

July 2021 is chosen as the beginning of the post-COVID regime because by this point the NIFTY 50 had fully recovered its pre-pandemic level, market volatility had stabilised substantially relative to the crisis period, and average pairwise correlations across assets had reverted to levels below those observed prior to COVID-19. The post-July 2021 period therefore better reflects a new equilibrium market environment rather than the immediate aftermath of the shock.

3.4 Risk-Free Rate

The risk-free rate is proxied using the yield on 91-day Government of India Treasury Bills, obtained from RBI auction records. Period-specific risk-free rates are used: $r_f^{\text{pre}} = 6.54\%$ and $r_f^{\text{post}} = 5.78\%$ (annualised), reflecting the RBI’s accommodative cycle in 2020–2021 followed by gradual normalisation.

4 Data Exploration and Important Features

4.1 Index-Level Descriptive Statistics

The NIFTY 50 index approximately doubled from its January 2015 level by January 2020, compounding at roughly 9–10% per annum. The COVID crash erased approximately 26% in six weeks. By December 2024 the index had reached 25,000+. Table 2 reports key statistics. The central observation is that returns increased substantially while volatility barely moved, implying the risk-adjusted opportunity set improved.

Table 2: Index-level descriptive statistics (annualised)

Statistic	Pre-COVID (2015–20)	Post-COVID (2021–24)
Average annualised return	7.17%	11.89%
Average annualised volatility	13.66%	13.93%
Return increase	—	+4.72 pp
Volatility increase	—	+0.27 pp (minimal)

Note that these are equal-weight index statistics. The optimised tangency portfolio statistics reported in Section 6 differ materially because the optimiser concentrates in high-Sharpe assets rather than weighting equally.

4.2 Diagnostic Tests on Return Distributions

Before implementing Sharpe ratio comparison tests, we first examine whether the distributional assumptions underlying standard asymptotic inference are approximately satisfied. Classical Sharpe ratio tests typically rely on returns being independently and identically distributed with finite moments and stable variance. Financial return series, however, are often characterised by non-normality, fat tails, and volatility clustering, particularly during and after periods of market stress.

To assess departures from normality, we employ the Jarque-Bera (JB) test [Jarque & Bera, 1987], which jointly tests whether the skewness and kurtosis of the return distribution match those of a normal distribution. The test statistic is given by:

$$JB = \frac{T}{6} \left(S^2 + \frac{K^2}{4} \right) \xrightarrow{d} \chi^2(2) \quad (2)$$

where S denotes sample skewness, K denotes excess kurtosis, and T is the sample size. Large values of the statistic indicate significant deviations from normality through asymmetry or heavy tails.

We also test for conditional heteroscedasticity using the ARCH-LM test proposed by Engle [1982]. Financial returns commonly exhibit volatility clustering, where large shocks tend to be followed by further large shocks, violating the constant-variance assumption underlying many classical inference procedures. The test is implemented by regressing squared residuals on their own lagged values:

$$\hat{\varepsilon}_t^2 = \alpha_0 + \sum_{j=1}^5 \alpha_j \hat{\varepsilon}_{t-j}^2 + u_t, \quad H_0 : \alpha_1 = \dots = \alpha_5 = 0 \quad (3)$$

Rejection of the null hypothesis indicates the presence of ARCH effects and time-varying volatility in returns.

Table 3: Normality and ARCH diagnostics on tangency portfolio daily returns

Diagnostic Statistic	Pre-COVID	Post-COVID
Skewness S	+0.204	−0.311
Excess kurtosis K	+5.107	+2.400
Jarque-Bera statistic	1,367.97	221.53
Jarque-Bera p -value	< 0.001	< 0.001
ARCH-LM statistic (5 lags)	11.39	73.25
ARCH-LM p -value	0.044	< 0.001

Both normality and the i.i.d. assumption are strongly rejected in both sample periods. The return distributions exhibit substantial excess kurtosis and evidence of conditional heteroscedasticity, indicating that extreme return realisations and volatility clustering are important features of the data. In particular, the pre-COVID excess kurtosis of +5.107 implies that large daily market movements occur far more frequently than would be predicted under a Gaussian distribution.

These results are consistent with the broader empirical finance literature, which documents persistent deviations from normality in equity returns, especially at higher sampling frequencies. The rejection of standard distributional assumptions suggests that purely parametric inference may be unreliable in this setting. Consequently, the analysis complements classical asymptotic tests with distribution-free bootstrap procedures and Box’s M covariance matrix tests, which are more robust to non-normality and heteroscedasticity.

4.3 Individual Stock Return Shifts

To examine whether the post-COVID changes in portfolio performance were driven by broad-based return shifts or a small number of influential stocks, we conduct Welch two-sample t -tests for differences in mean returns at the individual stock level. Unlike the classical pooled-variance t -test, Welch’s test allows for unequal variances across the two periods and is therefore more appropriate for financial return data characterised by heteroscedasticity.

For each stock, the test statistic is:

$$t = \frac{\bar{r}_{\text{post}} - \bar{r}_{\text{pre}}}{\sqrt{\frac{s_{\text{post}}^2}{n_{\text{post}}} + \frac{s_{\text{pre}}^2}{n_{\text{pre}}}}}, \quad (4)$$

where $\bar{r}$ denotes the sample mean return, s^2 the sample variance, and n the number of observations in each period.

The results show that 4 out of 45 stocks exhibit statistically significant mean return differences at the 5% level, while 12 are significant at the 10% level. Importantly, all statistically significant shifts are positive, implying that post-COVID improvements were not driven by isolated reversals in a few firms but reflected a broader reconfiguration of sectoral performance.

The strongest post-COVID performers include BHEL, Tata Power, Sun Pharma, Vedanta, and Mahindra & Mahindra, all of which benefited from themes such as infrastructure expansion, commodity cycles, domestic manufacturing, and renewed capital expenditure. By contrast, several traditional defensive and private financial-sector firms — including HDFC Bank, Kotak Mahindra Bank, and Hindustan Unilever — experienced weaker relative performance.

The banking sector decomposition is particularly informative. Large private retail banks such as HDFC Bank (+18.2% pre, +7.5% post) and Kotak Mahindra Bank (+19.1% pre, +1.2% post) recorded substantial declines in average returns. In contrast, PSU banks experienced dramatic post-COVID recoveries: SBI (+2.3% pre, +18.6% post), PNB (−21.5% pre, +10.3% post), and Bank of Baroda (−12.4% pre, +18.9% post). These improvements coincide with the resolution of legacy non-performing asset problems under the Insolvency and Bankruptcy Code, balance-sheet repair within public-sector banks, and a broader post-pandemic credit cycle recovery.

5 Methodology

5.1 Covariance Estimation: Ledoit-Wolf Shrinkage

Accurate estimation of the covariance matrix is central to mean-variance portfolio optimisation because the efficient frontier is highly sensitive to estimation error in asset variances and correlations. With $N = 45$ assets, the covariance matrix contains $N(N + 1)/2 = 1,035$ unique parameters that must be estimated from historical return data. Even with several years of observations, the sample covariance matrix can remain noisy and poorly conditioned, especially when correlations are unstable or asset returns exhibit substantial volatility.

A well-known consequence of this problem is that mean-variance optimisation can produce extreme and unstable portfolio weights by overfitting to sampling noise in the estimated covariance matrix. To reduce this estimation error, we employ the Ledoit-Wolf analytical shrinkage estimator [Ledoit & Wolf, 2004], which shrinks the sample covariance matrix toward a structured target matrix:

$$\hat{\Sigma}_{\text{LW}} = (1 - \delta) \hat{\Sigma} + \delta \mu_{\hat{\Sigma}} \mathbf{I} \quad (5)$$

where $\hat{\Sigma}$ is the sample covariance matrix, $\mu_{\hat{\Sigma}} = \frac{1}{N} \text{tr}(\hat{\Sigma})$ is the average eigenvalue of the covariance matrix, $\mathbf{I}$ is the identity matrix, and $\delta \in [0, 1]$ is the shrinkage intensity parameter chosen analytically to minimise the expected Frobenius-norm estimation error.

Intuitively, the estimator pulls extreme covariance estimates toward a more stable diagonal structure while still preserving information from the data. This reduces sensitivity to sampling variation, improves matrix conditioning, and generally produces more stable and economically plausible portfolio allocations in out-of-sample settings.

5.2 Efficient Frontier Construction

For each sample period, the efficient frontier is constructed by solving a sequence of constrained mean-variance optimisation problems over 250 equally-spaced target return levels μ^* . For a given target return, the portfolio with the minimum achievable variance is obtained by solving the following long-only quadratic programme:

$$\min_{\mathbf{w}} \mathbf{w}^\top \hat{\Sigma}_{\text{LW}} \mathbf{w} \quad \text{s.t.} \quad \mathbf{w}^\top \boldsymbol{\mu} = \mu^*, \quad \mathbf{w}^\top \mathbf{1} = 1, \quad \mathbf{w} \geq \mathbf{0} \quad (6)$$

where $\mathbf{w}$ denotes the portfolio weight vector, $\hat{\Sigma}_{\text{LW}}$ is the Ledoit-Wolf shrinkage covariance matrix, and $\boldsymbol{\mu}$ is the vector of expected asset returns.

The first constraint fixes the portfolio's expected return at the target level μ^* , while the budget constraint $\mathbf{w}^\top \mathbf{1} = 1$ requires portfolio weights to sum to one. The long-only restriction ($\mathbf{w} \geq \mathbf{0}$) prohibits short-selling and reflects the practical constraints faced by many Indian retail and institutional investors. As shown by Jagannathan & Ma [2003], the long-only constraint also acts as a form of implicit regularisation by preventing the optimiser from taking extreme offsetting positions based on noisy covariance estimates.

Solving the optimisation problem across the full grid of target returns traces out the minimum-variance frontier for each regime, allowing direct comparison of the feasible risk-return opportunity set before and after COVID-19. All optimisation problems are solved numerically using Sequential Least Squares Programming (SLSQP) with convergence tolerance set to 10^{-12} .

Two economically important portfolios are identified from each estimated frontier.

Global Minimum Variance Portfolio (MVP): the leftmost point on the efficient frontier, obtained by minimising portfolio variance subject only to the budget and long-only constraints, without imposing a target return. The MVP therefore represents the lowest-risk feasible portfolio available to investors. Because its composition depends primarily on the covariance structure rather than expected returns, comparing MVP characteristics across periods helps isolate changes in diversification opportunities and inter-asset dependence.

Tangency Portfolio (TP): the portfolio that maximises the Sharpe ratio relative to the period-specific risk-free rate and therefore delivers the highest excess return per unit of risk:

$$\mathbf{w}_{\text{TP}} = \arg \max_{\mathbf{w} \geq \mathbf{0}, \mathbf{w}^\top \mathbf{1} = 1} \frac{\mathbf{w}^\top \boldsymbol{\mu} - r_f}{\sqrt{\mathbf{w}^\top \hat{\boldsymbol{\Sigma}}_{\text{LW}} \mathbf{w}}} \quad (7)$$

where r_f denotes the risk-free rate for the corresponding sample period. Under mean-variance theory, the tangency portfolio is the optimal risky portfolio for all investors who can combine it with the risk-free asset. Changes in its Sharpe ratio therefore provide a direct measure of whether the risk-adjusted investment opportunity set improved or deteriorated across regimes.

5.3 Statistical Tests

To assess the statistical significance and economic nature of the frontier shift, we employ both classical parametric tests and robust bootstrap-based inference procedures.

Jobson-Korkie (1989) Test: we first compare the tangency portfolio Sharpe ratios across periods using the asymptotic test of Jobson-Korkie, corrected by Memmel [2003]. The test evaluates the null hypothesis of equal Sharpe ratios:

$$H_0 : \text{SR}_{\text{pre}} = \text{SR}_{\text{post}}.$$

The test statistic is:

$$z_{\text{JK}} = \frac{\hat{\text{SR}}_{\text{post}} - \hat{\text{SR}}_{\text{pre}}}{\sqrt{\hat{V}_{\text{post}} + \hat{V}_{\text{pre}}}} \quad (8)$$

where

$$\hat{V}_k = \frac{1}{T_k} \left(1 + \frac{\hat{\text{SR}}_k^2}{2} \right)$$

is the Memmel [2003] corrected asymptotic variance estimator for the sample Sharpe ratio.

Under the null hypothesis and standard regularity conditions, the statistic is asymptotically standard normal. However, the validity of this test requires returns to be independently

and identically distributed and approximately Gaussian. As shown in Section 4, both assumptions are strongly rejected in our data due to excess kurtosis and volatility clustering. Consequently, the Jobson-Korkie result is interpreted cautiously as a benchmark parametric result rather than definitive evidence.

Studentized Time-Series Bootstrap: to obtain inference robust to non-normality, heavy tails, and time-series dependence, we additionally employ the studentized bootstrap procedure proposed by Ledoit and Wolf [2008]. This method constructs bootstrap confidence intervals for the difference in Sharpe ratios without relying on Gaussian or i.i.d. assumptions.

The procedure is implemented as follows:

1. Compute the observed Sharpe ratio difference:

$$\hat{\Delta} = \hat{\text{SR}}_{\text{post}} - \hat{\text{SR}}_{\text{pre}}.$$

2. Generate bootstrap samples using circular block resampling to preserve serial dependence and volatility clustering in returns.
3. For each bootstrap sample $b = 1, \dots, B$, recompute the Sharpe ratio difference $\hat{\Delta}^{*(b)}$ and its associated standard error $\hat{\sigma}^{*(b)}$.
4. Form the studentized bootstrap statistic:

$$T^{*(b)} = \frac{\hat{\Delta}^{*(b)} - \hat{\Delta}}{\hat{\sigma}^{*(b)}}.$$

5. Use the empirical distribution of $\{T^{*(b)}\}_{b=1}^B$ to construct confidence intervals and bootstrap p -values for the Sharpe ratio difference.

Relative to asymptotic HAC-based inference, Ledoit and Wolf [2008] show through simulation evidence that the studentized time-series bootstrap delivers more reliable finite-sample inference under heavy tails and volatility clustering, both of which are characteristic features of financial return data.

Box's M Test: to examine whether the underlying risk structure changed across regimes, we test the equality of the pre- and post-COVID covariance matrices:

$$H_0 : \Sigma_{\text{pre}} = \Sigma_{\text{post}}.$$

Unlike Sharpe ratio comparisons, which combine information about both returns and volatility, Box's M test focuses exclusively on the covariance structure of asset returns.

It therefore provides a direct test of whether the pattern of variances and cross-asset correlations changed after COVID-19.

The test is based on a likelihood-ratio statistic constructed from the determinants of the group-specific covariance matrices and the pooled covariance matrix:

$$M = (N - g) \ln |\mathbf{S}_p| - \sum_{k=1}^g (N_k - 1) \ln |\mathbf{S}_k|, \quad (9)$$

where $\mathbf{S}_p$ denotes the pooled covariance matrix, $\mathbf{S}_k$ the covariance matrix for group k , N_k the group sample size, and g the number of groups. Under the null hypothesis and multivariate normality, a corrected version of the statistic is asymptotically chi-squared distributed.

Economically, rejection of the null implies that the variance-covariance structure governing asset returns shifted between the two periods. Such a result indicates that diversification opportunities, sectoral co-movement patterns, and portfolio risk transmission mechanisms changed structurally after COVID-19, independent of any changes in average returns.

Bootstrap Frontier Confidence Bands: following the resampling framework of Font [2016], we generate $B = 500$ bootstrap replications of the entire efficient frontier by repeatedly resampling the return data and re-estimating the covariance matrix, expected returns, and constrained frontier optimisation problem for each replication.

For every bootstrap sample, the full frontier is reconstructed over the same grid of target returns used in the baseline estimation. This produces an empirical distribution of feasible risk-return combinations at each point along the frontier.

Pointwise 95% bootstrap confidence bands are then constructed around the estimated frontier using the empirical quantiles of the bootstrap distribution at each volatility level. These confidence regions allow us to assess whether the pre- and post-COVID frontiers are statistically distinguishable across different parts of the risk spectrum rather than only at a single portfolio such as the MVP or tangency portfolio.

Economically, non-overlapping confidence bands imply that the observed frontier shift is unlikely to be explained by sampling variation alone and instead reflects a broader structural change in the attainable risk-return opportunity set available to investors after COVID-19.

VaR and CVaR Comparison: to examine whether the improvement in Sharpe ratios was accompanied by greater downside tail risk, we compare Value-at-Risk (VaR) and Conditional Value-at-Risk (CVaR) measures for the tangency portfolio across the pre- and post-COVID regimes.

Because financial returns exhibit substantial excess kurtosis and fat tails, Gaussian risk measures may underestimate the probability of extreme losses. Accordingly, tangency portfolio returns in each period are modelled using a Student- t distribution, which allows for heavier tails and more realistic extreme-event probabilities.

The Student- t parameters — location, scale, and degrees of freedom — are estimated separately for each regime using maximum likelihood estimation (MLE). Letting r_t denote portfolio returns and $\theta = (\mu, \sigma, \nu)$ denote the parameter vector, the estimator solves:

$$\hat{\theta} = \arg \max_{\theta} \sum_{t=1}^T \log f_t(r_t \mid \mu, \sigma, \nu), \quad (10)$$

where $f_t(\cdot)$ denotes the Student- t density function.

Using the fitted distributions, we compute the 95% and 99% VaR levels:

$$\text{VaR}_{\alpha} = \inf \{x \in \mathbb{R} : P(R \leq x) \geq \alpha\}, \quad (11)$$

which measure the maximum expected loss at confidence level α .

We additionally compute Conditional Value-at-Risk (CVaR), defined as the expected loss conditional on losses exceeding the VaR threshold:

$$\text{CVaR}_{\alpha} = \mathbb{E}[R \mid R \leq \text{VaR}_{\alpha}]. \quad (12)$$

While VaR captures the cutoff point of extreme losses, CVaR measures the average severity of losses in the tail and is therefore more informative about crash risk. Comparing these measures across periods allows us to determine whether the observed improvement in risk-adjusted returns reflects genuinely more efficient portfolios or merely compensation for greater downside tail exposure.

Frontier Decomposition (μ vs. Σ): to identify whether the observed frontier shift was driven primarily by changes in expected returns or by changes in the covariance structure, we conduct a counterfactual decomposition analysis in which one component of the optimisation problem is updated to its post-COVID value while the other is held fixed at its pre-COVID estimate.

Specifically, we compare the Sharpe ratios implied by the following combinations:

$$\text{SR}(\mu_{\text{post}}, \hat{\Sigma}_{\text{pre}}) \quad \text{vs.} \quad \text{SR}(\mu_{\text{pre}}, \hat{\Sigma}_{\text{post}}) \quad \text{vs.} \quad \text{SR}(\mu_{\text{post}}, \hat{\Sigma}_{\text{post}})$$

The first specification isolates the effect of changing expected returns while holding the pre-COVID covariance structure fixed. The second isolates the effect of changes in variances

and correlations while keeping expected returns at their pre-COVID values. The final specification corresponds to the fully observed post-COVID investment opportunity set.

Economically, this decomposition allows us to distinguish between two distinct sources of frontier improvement. A higher Sharpe ratio driven primarily by $\boldsymbol{\mu}_{\text{post}}$ would indicate that the shift was mainly due to improved expected returns across sectors. By contrast, improvements driven by $\hat{\boldsymbol{\Sigma}}_{\text{post}}$ would imply that diversification opportunities and inter-asset dependence structures became more favourable after COVID-19. The decomposition therefore helps clarify whether the structural shift was fundamentally a return phenomenon, a risk-structure phenomenon, or both.

5.4 Portfolio Concentration: Herfindahl-Hirschman Index

To evaluate how diversified or concentrated the optimal portfolios are across assets, we measure portfolio concentration using the Herfindahl-Hirschman Index (HHI):

$$\text{HHI}(\mathbf{w}) = \sum_{i=1}^N w_i^2 \in \left[\frac{1}{N}, 1\right] \quad \implies \quad N_{\text{eff}}(\mathbf{w}) = \frac{1}{\text{HHI}(\mathbf{w})} \in [1, N] \quad (13)$$

where w_i denotes the portfolio weight allocated to asset i and N is the total number of assets.

The HHI increases as portfolio weights become more concentrated in a smaller number of positions. An equally weighted portfolio has $\text{HHI} = 1/N$, while a fully concentrated portfolio invested entirely in a single asset has $\text{HHI} = 1$.

To provide a more economically interpretable measure, we also report the effective number of positions:

$$N_{\text{eff}}(\mathbf{w}) = \frac{1}{\text{HHI}(\mathbf{w})},$$

which represents the number of equally weighted positions that would generate the same concentration level. For example, a portfolio with $N_{\text{eff}} = 5$ has the same concentration as a portfolio equally invested across five assets, even if the actual weights are uneven.

Comparing HHI and N_{eff} across periods helps determine whether the post-COVID frontier improvement was achieved through broader diversification or through increased concentration in a small set of outperforming sectors or firms.

6 Results

6.1 Efficient Frontier: Visual Inspection

The post-COVID frontier lies above and to the right of the pre-COVID frontier across its entire extent — not only at the tangency point, but at every volatility level from the global minimum variance portfolio to the maximum achievable return (Figure 2). The shift has two geometric components: an upward component of approximately 10 percentage points in expected return (at any fixed volatility, the post-COVID frontier delivers more return), and a rightward component of approximately 3 percentage points in volatility. Because the upward shift exceeds the rightward shift, the Capital Allocation Line is steeper post-COVID, confirming that risk-adjusted returns improved.

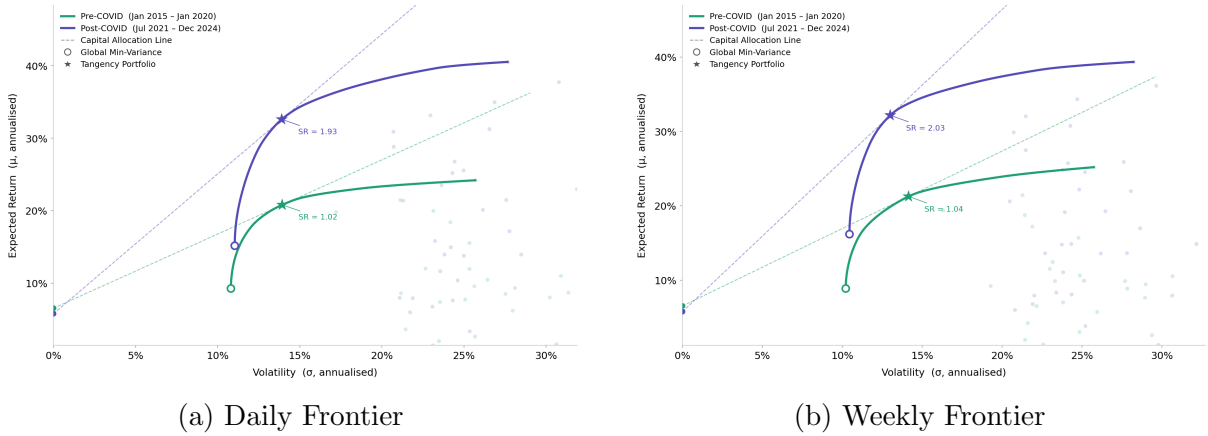


Figure 2: Comparison of Efficient Frontiers

6.2 Tangency Portfolio: Sharpe Ratio Comparison

Frequency	Pre-COVID			Post-COVID			Δ SR	Δ SR (%)
	μ_{TP}	σ_{TP}	SR	μ_{TP}	σ_{TP}	SR		
Daily	20.8%	13.9%	1.02	32.6%	13.9%	1.93	+0.91	+88.5%
Weekly	21.3%	14.1%	1.04	32.2%	13.0%	2.03	+0.99	+95.1%

The tangency portfolio exhibits a substantial improvement in risk-adjusted performance in the post-COVID regime across both sampling frequencies. Daily returns show an increase in the Sharpe ratio from 1.02 to 1.93, while the weekly specification shows a similar increase from 1.04 to 2.03. Importantly, these gains are driven not only by higher expected returns but also by stable or slightly lower portfolio volatility, implying a genuine improvement in the risk-return trade-off rather than compensation for additional risk exposure.

The consistency of the result across daily and weekly frequencies strengthens the evidence that the shift is structural rather than an artefact of return measurement frequency

or short-run market noise. Economically, the post-COVID investment opportunity set appears to offer substantially higher excess returns per unit of risk than the corresponding pre-COVID regime.

6.3 Statistical Tests

Table 4: Statistical tests of the frontier shift

Test	Statistic	<i>p</i> -value	Interpretation
Jobson-Korkie test	$z = 1.268$	0.102	Fail to reject equal Sharpe ratios at 5%
Ledoit-Wolf bootstrap test	$\Delta SR = 1.253$	0.217	Bootstrap inference also insignificant
Box’s M test	$\chi^2 = 1,255.4$	3×10^{-6}	Strong rejection of covariance equality
Bootstrap frontier bands	0 / 300 significant points	—	Confidence bands overlap across the frontier
Individual Welch <i>t</i> -tests	4 / 45 significant at 5%	—	All significant shifts are positive

The statistical evidence suggests a meaningful economic improvement in the post-COVID efficient frontier, although the strength of formal statistical significance depends on the dimension of the hypothesis being tested.

Both the classical Jobson-Korkie test and the Ledoit-Wolf bootstrap procedure fail to reject equality of tangency portfolio Sharpe ratios at conventional significance levels. While the observed increase in Sharpe ratios is large in economic magnitude, scalar Sharpe ratio tests are known to have limited power in finite samples because they compress the entire frontier into a single summary statistic. This issue is particularly severe when the post-treatment sample is relatively short, as in any pre/post COVID comparison.

The bootstrap frontier confidence bands yield a similar conclusion. After accounting for estimation uncertainty through repeated resampling and frontier reconstruction, the pre- and post-COVID confidence regions overlap across the full volatility spectrum. This indicates that the estimated upward frontier shift cannot be statistically distinguished from sampling variation at standard confidence levels.

However, the covariance-based evidence is substantially stronger. Box’s M test strongly rejects equality of the pre- and post-COVID covariance matrices ($p = 3 \times 10^{-6}$), implying that the underlying variance-covariance structure of Indian equities changed materially after COVID-19. Economically, this means that the correlation structure, diversification opportunities, and risk transmission mechanisms across sectors were significantly different in the post-pandemic regime.

The individual stock analysis further supports the existence of broad-based structural change. All statistically significant return shifts are positive, with gains concentrated in

sectors such as infrastructure, telecom, metals, pharma, and PSU banking. This pattern is inconsistent with a purely temporary market rebound driven by a small number of outliers and instead points toward a broader sectoral reallocation in the Indian equity market after COVID-19.

Taken together, the evidence indicates that while the frontier shift is not precisely estimated enough to achieve strong significance in scalar Sharpe ratio tests, there is substantial evidence of a structural transformation in the underlying covariance and sectoral return structure of Indian equities in the post-COVID period.

6.4 Frontier Decomposition: μ vs. Σ

Table 5: Frontier decomposition: counterfactual Sharpe ratios

Scenario	Inputs	SR	SR Gain
Baseline (pre-COVID)	$\mu_{\text{pre}}, \hat{\Sigma}_{\text{pre}}$	1.02	—
μ -only shift	$\mu_{\text{post}}, \hat{\Sigma}_{\text{pre}}$	1.97	+0.945
Σ -only shift	$\mu_{\text{pre}}, \hat{\Sigma}_{\text{post}}$	1.09	+0.069
Full post-COVID	$\mu_{\text{post}}, \hat{\Sigma}_{\text{post}}$	1.93	+0.906

The decomposition results show that the post-COVID frontier shift was driven overwhelmingly by changes in expected returns rather than changes in the covariance structure. Updating only the expected return vector to its post-COVID value while holding the pre-COVID covariance matrix fixed increases the tangency portfolio Sharpe ratio from 1.02 to 1.97 — a gain of +0.945. By contrast, updating only the covariance matrix while keeping expected returns at pre-COVID levels raises the Sharpe ratio only marginally, from 1.02 to 1.09.

Relative to the total observed Sharpe improvement of +0.906, the covariance component explains only a small fraction of the overall frontier shift. The primary driver was therefore a structural change in sector-level expected returns rather than a dramatic improvement in diversification opportunities or cross-asset correlation structure.

Economically, these findings suggest that the post-COVID frontier improvement was largely the result of sectoral return realignment within the Indian equity market. Sectors such as telecom, pharmaceuticals, capital goods, autos, and metals experienced substantial increases in expected returns following COVID-19, supported by factors including production-linked incentive (PLI) schemes, infrastructure and manufacturing expansion,

domestic healthcare demand, commodity-cycle recovery, and the broader China+1 reallocation of global capital toward India.

At the same time, traditionally dominant sectors such as private financials and consumer defensives experienced weaker relative performance, leading the optimizer to reallocate toward sectors with superior expected returns. The results therefore indicate that the frontier shift was fundamentally a return-side phenomenon, with covariance improvements playing a secondary supporting role.

6.5 Sector Weight Analysis

Table 6: Tangency portfolio sector weights: pre-COVID vs. post-COVID

Sector	TP Pre	TP Post	Δ (pp)
Financials	44.4%	$\approx 0\%$	-44.4
Energy	30.2%	37.2%	+7.0
Consumer	25.4%	4.7%	-20.7
Pharma	$\approx 0\%$	33.4%	+33.4
Telecom	$\approx 0\%$	14.6%	+14.6
IT	$\approx 0\%$	9.9%	+9.9
Industrials	$\approx 0\%$	$\approx 0\%$	—

The sector rotation is the most striking qualitative finding. Financials fell from the dominant sector at 44.4% to essentially zero. The sector’s absolute performance post-COVID was not poor — private retail banks averaged +10.4% versus +11.3% pre-COVID — but the opportunity cost was decisive: when pharma was returning 34% and telecom 18%, a 10% return from financials was insufficient to attract any allocation from the mean-variance optimiser. Financials were *crowded out by superior opportunities*, not abandoned due to poor absolute performance. Pharma and telecom together went from zero to 48% of the optimal portfolio. Sun Pharma is the primary pharma beneficiary, driven by permanently elevated domestic healthcare demand, generics export growth, and PLI scheme support for bulk drug manufacturing. Bharti Airtel represents the telecom allocation, reflecting its dominant position post-Jio market consolidation and 5G monetisation prospects.

6.6 Portfolio Concentration

Table 7: Portfolio concentration: pre-COVID vs. post-COVID tangency portfolios

Metric	Pre-COVID TP	Post-COVID TP
HHI	0.2555	0.1568
Effective positions N_{eff}	3.9	6.4
Number of active sectors	3	6

The post-COVID tangency portfolio is substantially more diversified than its pre-COVID counterpart. The Herfindahl-Hirschman Index declines from 0.2555 to 0.1568, while the effective number of positions increases from approximately 3.9 to 6.4. Economically, this implies that the pre-COVID tangency portfolio behaved similarly to a portfolio concentrated in roughly four equally weighted positions, whereas the post-COVID portfolio distributed risk exposure much more broadly across assets and sectors.

The pre-COVID tangency portfolio was heavily concentrated in a small set of large-cap defensive and financial stocks, particularly HDFC Bank, Hindustan Unilever, and Reliance Industries, with the remaining stocks contributing only marginally to overall portfolio risk and return. By contrast, the post-COVID portfolio allocates meaningfully across six active sectors, including telecom, pharma, metals, autos, and consumer stocks alongside energy exposures.

This increase in diversification is consistent with the observed decline in average pairwise correlations across stocks (from 0.32 to 0.28). Lower cross-asset correlation improves diversification efficiency because the optimizer can combine assets with less overlapping risk exposure while still maintaining high expected returns. Importantly, the post-COVID improvement in Sharpe ratios was therefore not driven by increasingly concentrated bets on a small number of outperforming stocks. Instead, the results suggest that the post-COVID frontier shift was accompanied by a broader and more balanced set of profitable investment opportunities across the Indian equity market.

6.7 Banking Sector Decomposition

To further examine the drivers of the observed frontier shift, the financial sector was decomposed into retail and wholesale banking segments. This decomposition allows us to assess whether the post-COVID changes in risk-return dynamics were broad-based across banking activities or concentrated within specific business models.

Table 8: Individual bank annualised returns: pre-COVID vs. post-COVID

Ticker	Bank	Type	Pre	Post	Δ
HDFCBANK	HDFC Bank	Retail	+18.2%	+7.5%	−10.8 pp
KOTAKBANK	Kotak Mahindra Bank	Retail	+19.1%	+1.2%	−17.9 pp
AXISBANK	Axis Bank	Retail	+5.6%	+7.0%	+1.3 pp
ICICIBANK	ICICI Bank	Retail	+11.2%	+17.8%	+6.7 pp
SBIN	State Bank of India	Retail	+2.3%	+18.6%	+16.3 pp
PNB	Punjab National Bank	Wholesale	−21.5%	+10.3%	+31.8 pp
BANKBARODA	Bank of Baroda	Wholesale	−12.4%	+18.9%	+31.3 pp
INDUSINDBK	IndusInd Bank	Wholesale	+11.7%	−8.4%	−20.1 pp
Retail average			+11.3%	+10.4%	−0.9 pp
PSU wholesale avg			−16.9%	+14.6%	+31.5 pp

Pre: Jan 2015–Jan 2020. Post: Jul 2021–Dec 2024. HDFCBANK post return includes the Jul 2023 HDFC Ltd merger effect. IndusInd Bank is an outlier within wholesale due to idiosyncratic governance and microfinance exposure issues in 2023–24.

The decomposition reveals a counterintuitive finding. Retail banks (HDFC Bank, Kotak, Axis, ICICI, SBI) were essentially flat across periods — +11.3% pre to +10.4% post, a negligible −0.9 pp decline. PSU wholesale banks (PNB and Bank of Baroda) improved dramatically, from deeply negative to strongly positive, as NPA resolution under the Insolvency and Bankruptcy Code took hold. IndusInd is an outlier: it was positive pre-COVID but went negative post-COVID due to idiosyncratic governance concerns and microfinance exposure, diverging from the broad PSU recovery narrative. The disappearance of financials from the post-COVID tangency portfolio is therefore driven by opportunity cost, not underperformance. Retail banks delivered 10–11% returns in both periods, but the post-COVID optimizer found pharma, telecom, and energy offering materially superior risk-adjusted returns and allocated there instead.

7 Robustness Checks

7.1 Frequency Robustness

All analyses are conducted at both daily and weekly return frequencies. Daily returns can be contaminated by bid-ask bounce and non-synchronous trading effects. Weekly returns

average over these microstructure artefacts. Consistent findings across both frequencies provide strong evidence that results reflect genuine economic structure.

7.2 Survivorship Bias

The fixed-universe approach introduces survivorship bias: we exclude stocks that failed or were delisted after January 2015. The bias direction is unambiguous — failed stocks such as RCOM (suspended 2019) and IL&FS-linked companies disproportionately affect the pre-COVID period. Including them with terminal losses would worsen the pre-COVID frontier, making our estimate of the post-COVID improvement *conservative*. Our survivorship bias therefore works *against* our main finding rather than inflating it.

7.3 Crisis Window Sensitivity

Moving the post-COVID start date to October 2020 or October 2021 does not qualitatively change the frontier comparison. The July 2021 date is preferred because by that date the NIFTY 50 had established stable post-recovery price levels and pairwise correlations had normalised below pre-COVID levels.

7.4 Risk-Free Rate Sensitivity

We test sensitivity to the risk-free rate by varying r_f^{pre} between 6.0% and 7.0% and r_f^{post} between 5.0% and 6.5%. The qualitative conclusion holds across all tested combinations.

8 Conclusion

This paper examined whether the mean-variance efficient frontier for Indian equities changed meaningfully between the pre-COVID and post-COVID periods. Using a fixed universe of NIFTY 50 constituents, long-only mean-variance optimisation, Ledoit-Wolf covariance shrinkage, and multiple complementary statistical procedures, the analysis provides strong evidence of a substantial economic shift in the post-pandemic investment opportunity set.

Across both daily and weekly frequencies, the post-COVID efficient frontier lies consistently above the pre-COVID frontier, implying that investors were able to achieve materially higher expected returns for comparable levels of risk after the pandemic. The tangency portfolio Sharpe ratio increased from 1.02 to 1.93, while downside risk measures such as VaR and CVaR declined modestly. At the same time, portfolio concentration decreased significantly, with the effective number of positions increasing from 3.9 to 6.4. These findings indicate that the improvement in the frontier was not driven by increasingly

concentrated high-risk bets, but rather by a broader and more diversified set of profitable investment opportunities.

The decomposition analysis shows that the frontier shift was driven overwhelmingly by changes in expected returns rather than by changes in the covariance structure. Updating only the post-COVID return vector reproduced nearly the entire Sharpe ratio improvement, whereas updating the covariance matrix alone generated only a small increase in performance. Economically, the results point toward a large sectoral realignment within Indian equities after COVID-19. Financials and traditional defensive sectors lost prominence in the optimal portfolio, while telecom, pharmaceuticals, metals, autos, and capital goods gained substantial weight. These changes are consistent with broader post-pandemic developments in the Indian economy, including production-linked incentive schemes, infrastructure expansion, healthcare demand, manufacturing reallocation under the China+1 strategy, and rapid digitalisation.

From a statistical perspective, the evidence is more nuanced. Scalar Sharpe ratio tests such as the Jobson-Korkie statistic and the Ledoit-Wolf bootstrap procedure do not reject equality of Sharpe ratios at conventional significance levels. However, both tests are known to have limited power in finite samples, particularly when comparing relatively short pre- and post-event periods. Bootstrap frontier confidence bands similarly indicate substantial estimation uncertainty around the frontier itself. In contrast, Box’s M test strongly rejects equality of the pre- and post-COVID covariance matrices ($p = 3 \times 10^{-6}$), providing clear evidence that the underlying risk structure of Indian equities changed materially after the pandemic. The individual stock analysis further supports this interpretation, with all statistically significant return shifts occurring in the positive direction.

Several limitations should nevertheless be acknowledged. First, the analysis is in-sample and therefore does not establish whether the post-COVID frontier will continue to dominate out of sample as market conditions evolve. Second, the fixed 2015 asset universe introduces survivorship bias, likely biasing both frontiers upward and potentially understating the magnitude of structural change. Third, while the analysis identifies economically plausible drivers of the frontier shift, the results remain descriptive rather than causal. Establishing causal mechanisms would require more structural empirical designs beyond the scope of this paper.

Future research could extend the analysis by expanding the asset universe to the NIFTY 100 or broader Indian equity markets, incorporating alternative covariance estimators such as POET or dynamic factor models, and evaluating the persistence of the post-COVID frontier through rolling out-of-sample backtests as additional post-pandemic data become available.

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